Multiple Imputation with Functional Data

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Introduction

What is imputation?

- Imputation methods replace missing values with statistically coherent estimates.
- Provides a way to analyze incomplete data using standard models.
- In functional data: sparsely observed curves can be viewed as partially missing functions.

Modern Multiple Imputation with Functional Data (Rao and Reimherr, 2020)

- Reviews existing imputation methods in the context of functional data analysis.
- Proposes modern multiple imputation strategies for sparse functional predictors.

Sparse Functional Data as a Missing Data Problem

- Latent functional process $X_i(t)$, with $t \in \mathcal{T}$
- Observations available only at subject-specific time points:

$$x_{ij} = X_i(t_{ij}) + \delta_{ij}, \quad j = 1, \ldots, m_i$$

- The number of observations m_i is small and varies across subjects
- Observations are irregularly sampled over $\mathcal{T}$
- When evaluated on a common grid, unobserved values are treated as missing data
- The main objective is therefore to reconstruct the latent curves $X_i(t)$ for use in subsequent functional regression analyses.

Why Multiple Imputation in Functional Regression

- Sparse functional data require reconstruction of the latent curves $X_i(t)$
- Single imputation produces one completed dataset:
  - ignores uncertainty in the reconstruction
  - may lead to biased estimation and underestimated variability
- Multiple imputation generates M completed datasets:

$$\{X_i^{(1)}(t), \ldots, X_i^{(M)}(t)\}$$

- Variability across imputations reflects uncertainty in missing parts of the curves
- Leads to improved inference in functional regression models

Baseline Method: PACE

Karhunen–Loève (KL) expansion $X_i(t) = \mu(t) + \sum_{k=1}^{\infty} \xi_{ik}\, v_k(t)$

1. Estimate mean and covariance from pooled sparse data: $\hat{\mu}(t),\ \hat{C}(s,t)$
2. Functional PCA of $\hat{C}$: obtain eigenfunctions $\hat{v}_k(t)$ and eigenvalues $\hat{\lambda}_k$
3. Truncate to K components and predict subject scores from observed points: $\hat{\xi}_{ik} = \mathbb{E}(\xi_{ik} \mid x_{i1}, \ldots, x_{im_i})$
4. Reconstruct a dense curve (single imputation): $\hat{X}_i(t) = \hat{\mu}(t) + \sum_{k=1}^{K} \hat{\xi}_{ik}\hat{v}_k(t)$
5. Under Gaussian assumptions, score prediction corresponds to a BLUP.

Limitations:

- Produces a single completed curve $\Rightarrow$ ignores imputation uncertainty.
- Imputation ignores the response Y $\Rightarrow$ potential bias in regression.

Baseline Method: MICE

- MICE is a general-purpose multiple imputation method for missing data. Functional data are discretized on a common grid: $X_i = (X_i(t_1), \ldots, X_i(t_p))$
- Each time point is treated as a variable with missing entries.
- Iterative procedure:
  - initialize missing values
  - sequentially fit models $X(t_j) \mid X(t_{-j})$
  - replace missing values with predicted draws
  - repeat for several cycles
- Repeating the algorithm yields M imputed datasets.

Limitations:

- Does not explicitly enforce smoothness of functional trajectories.
- High-dimensional when the grid is dense.

MissForest (MF)

1. Initialize missing values by mean imputation
2. Sort variables by increasing number of missing values
3. For each variable p:
   - fit random forest for observed $y_{obs}^{(p)}$ by means of observed $x_{obs}^{(p)}$
   - predict missing $y_{mis}^{(p)}$ by applying model to $x_{mis}^{(p)}$
4. Repeat imputation cycle until stop criteria is met

Advantages:

- No tuning parameter involved
- Works with any kind of data, especially with sparse data, too
- Models any kind of non-linearity, but **curves not smooth yet**

Local Linear Forest (LLF)

- Create random forest with B trees and leaves $L_b(x)$
- Determine weights

$$\alpha_i(x) = \frac{1}{B} \sum_{b=1}^{B} \frac{\mathbb{1}_{\{x_i \in L_b(x)\}}}{|L_b(x)|} \in [0, 1]$$

  as kernel for local linear regression

- Solve locally weighted least squares problem

$$\min_{\mu,\beta} \sum_{i=1}^{n} (Y_i - \mu - (x - x_i)^T \beta)^2 \, \alpha_i(x)$$

Newly implemented method: MLLF

Miss Local Linear Forest:

- Use basic principle of MF
- Instead of random forest, use LLF as model
- Benefit of most advantages from MF

Limitations:

- LLF does not automatically create multiple datasets
  $\Rightarrow$ manually create multiple imputed datasets and average
- E.g. through different initialization

Pace Initialization and Binning

- Curves produced by MF and MLLF not smooth yet
- To improve performance: **initialize imputations with PACE**

$\Rightarrow$ resulting methods denoted by MFP and MLLFP

- Computationally heavy, but smooths curve slightly
- To eventually smooth the curves: **use bins**

PACE Initialization and Binning

1. Divide m data points into k small intervalls (*bins*)
   - first bin: first data point
   - middle $k-2$ bins: divided equally, mean value of data in each bin used
   - last bin: last data point
2. Fit b-splines at the k bins
3. Interpolate back to the original m time points

- k parameter like bandwidth in kernel smoothing
- Also becomes less computationally heavy
- Resulting methods denoted by MF_B and MLLF_B

Empirical evaluation strategy

- The paper evaluates functional multiple-imputation methods using
  1. **a controlled simulation study** and
  2. **a real-data application** on Electronic Health Records (EHR)
- In both cases, the models are evaluated through a **scalar-on-function** regression framework

$$Y_i \;=\; \alpha + \int_{\mathcal{T}} X_i(t)\,\beta(t)\,dt + \varepsilon_i$$

- Performance is assessed through the Root Mean Square Error (RMSE) of the imputations, coefficient estimates and the predictions.

Simulation settings

- Functional predictors observed under **medium sparsity** (50%) and **high sparsity** (85%)
- Two response types considered: **continuous (scalar)** and **binary**
- Simulation is done from a Gaussian process with the Mátern covariance function.
- The simulations consider sample sizes $n \in \{200, 500, 1000\}$ and functional data observed on grids of size $m = 32$ and $m = 52$.

Simulation Study: Results

The main takeaways from the simulation study:

1. PACE and MICE underperform compared to the other models across all parameter settings.
2. Regardless of the sparsity, sample size and grid size, the binned methods **dominate** the other methods in terms of performance, with no clear winner between them.
3. MICE and MLLF perform relatively worse for high sparsity due a poor job of imputing the curves.
4. PACE initialization improves performance for both MF and MLLF, but the effect **disappears** as soon as we apply binning.

Simulation Study: Results

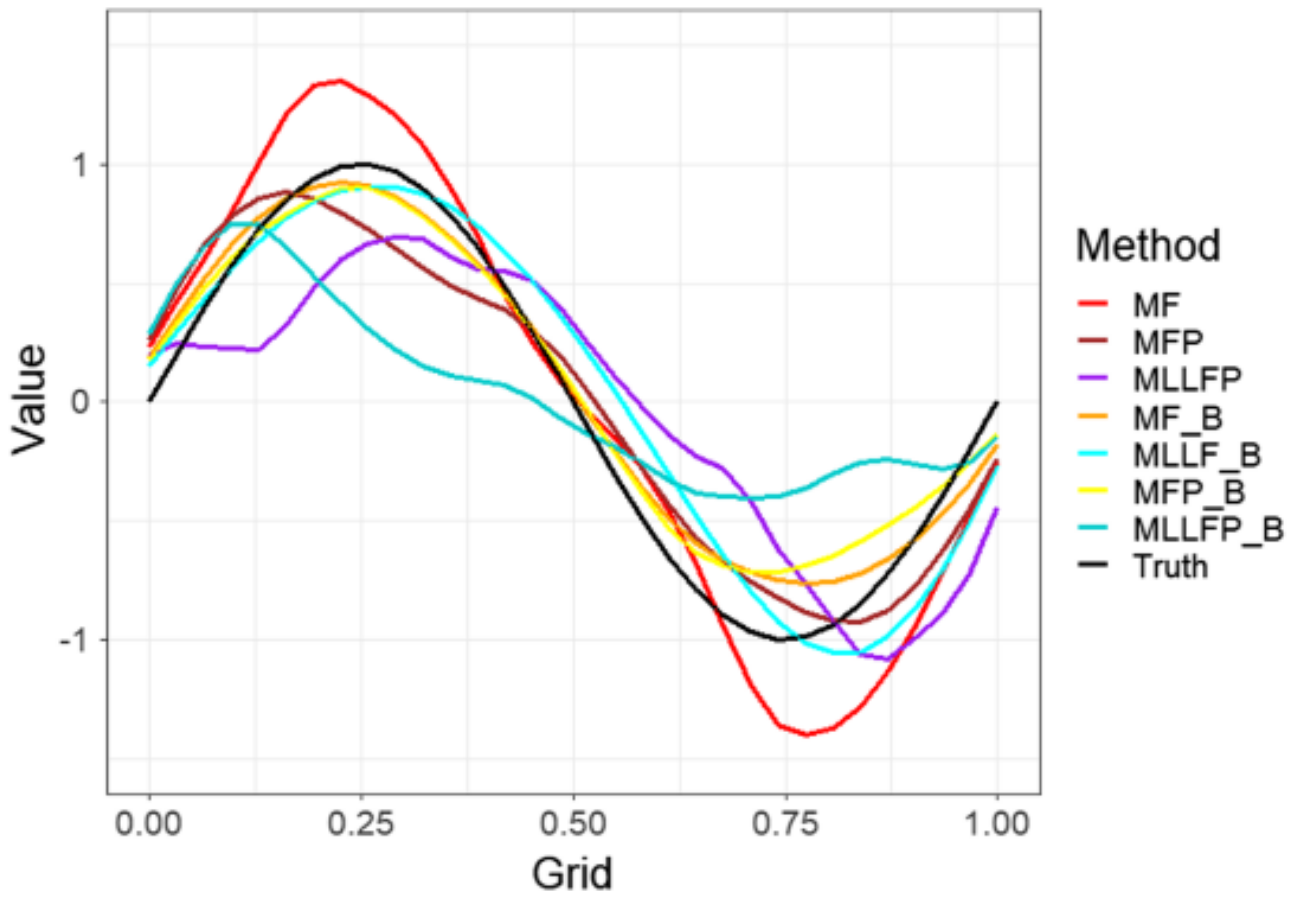

Figure: Estimated coefficient functions of the paper's simulation study (n = 500, m = 52, high sparsity)

Simulation Study: Results

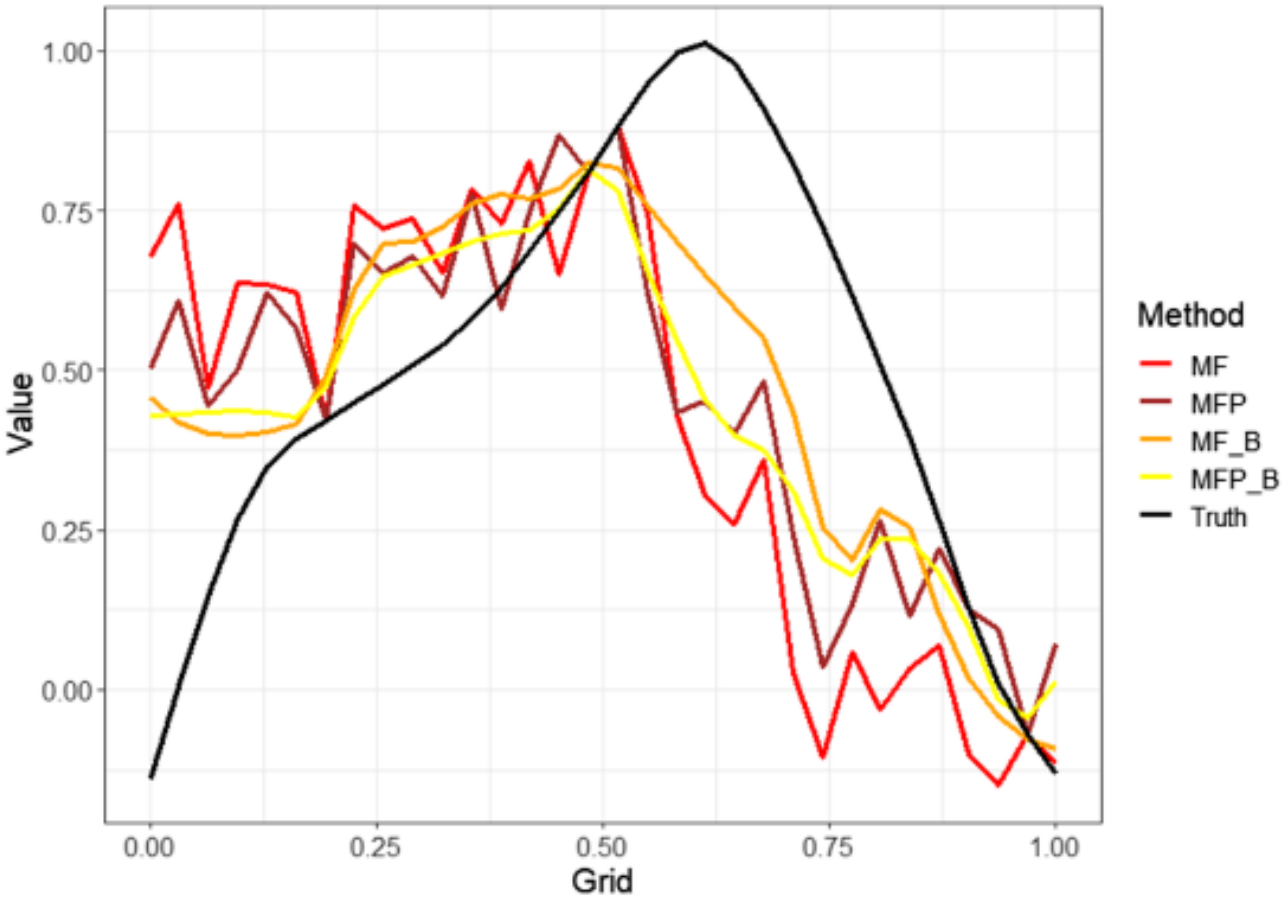

Figure: Comparing imputed curves in non-binned and binned methods of MF (m = 52, high sparsity)

EHR: Data

- The goal is to predict if a smoker will relapse based on their blood pressure recordings in the first 2 years of quitting.
- Due to the timing and the irregularity of doctor appointments, data is highly sparse.
- Number of observations per patient varies between 1 and 15, with 2 as the mode.
- Imputation choice affects inferential stability and uncertainty

EHR: Results

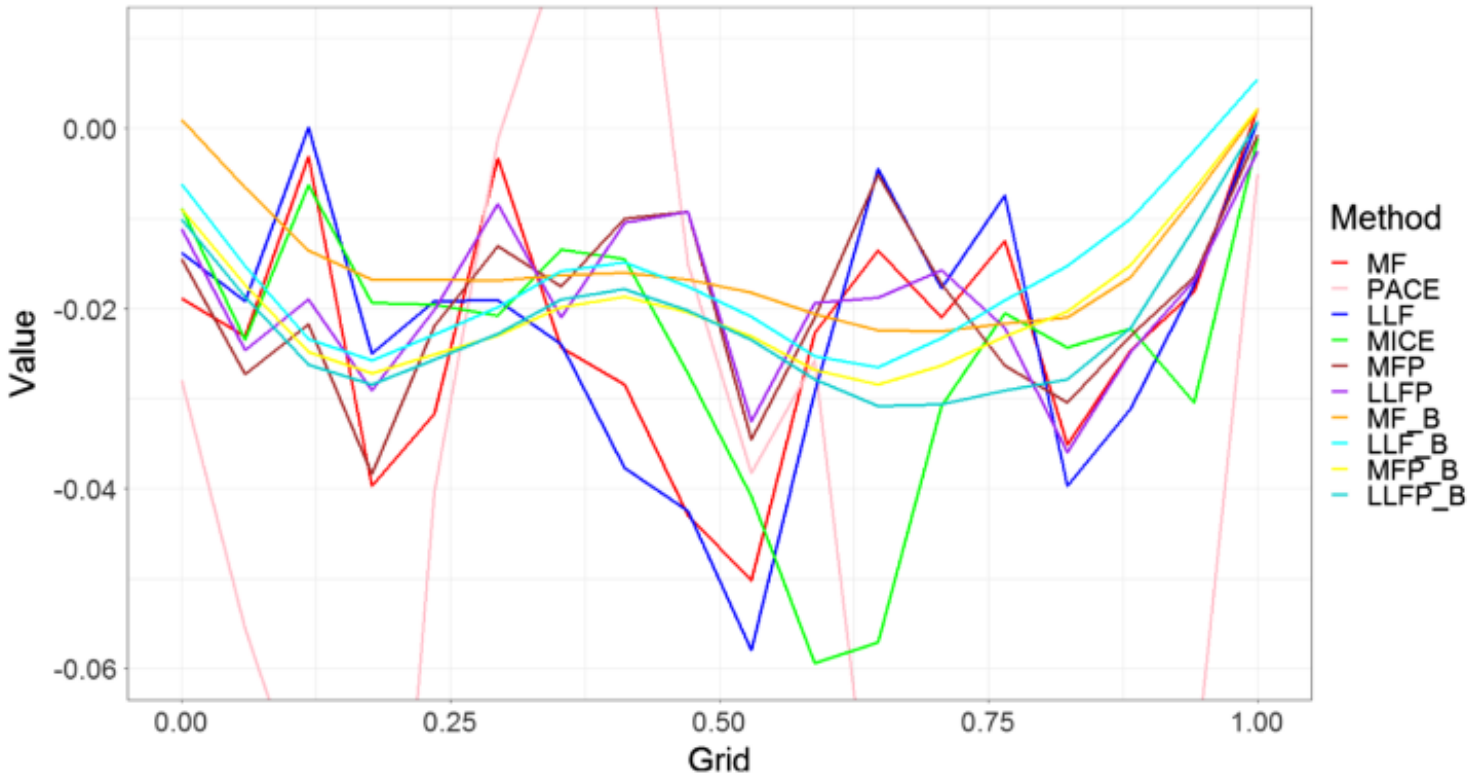

Figure: Estimated coefficient function of all the methods (n = 122, m = 18)

Application

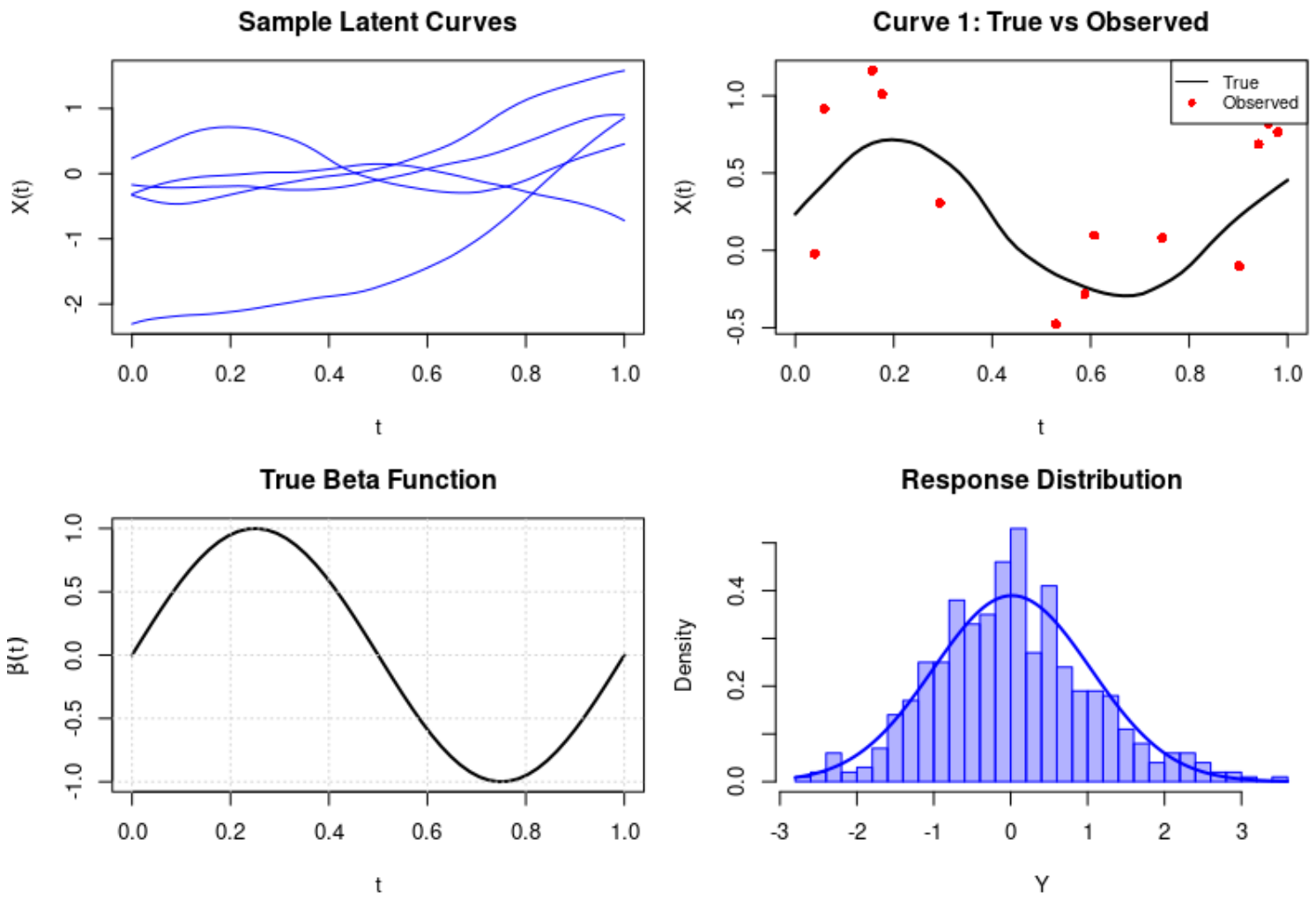

Figure: Simulated data according to the procedure in Rao and Reimherr, 2020

Application

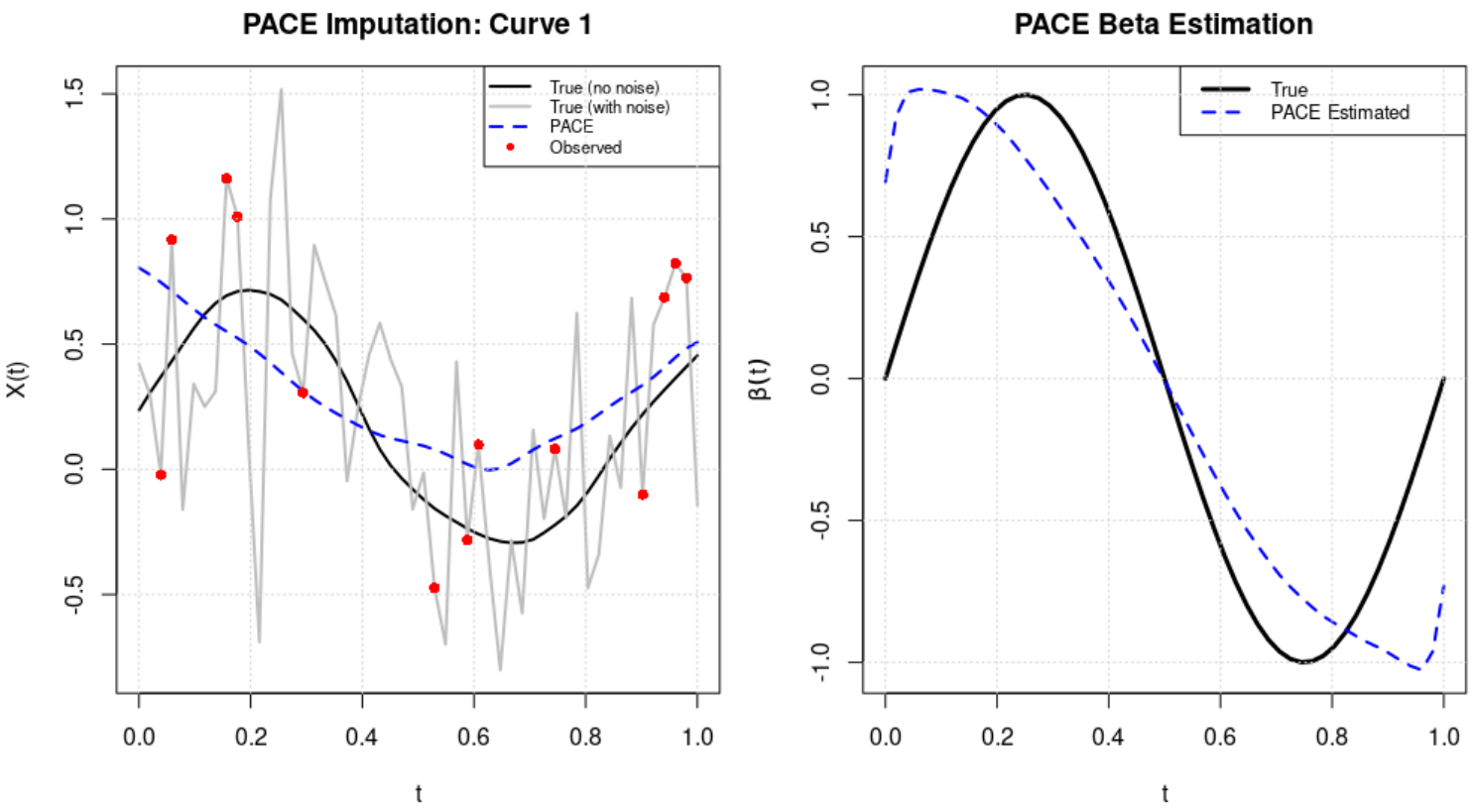

Figure: Results for PACE

Application

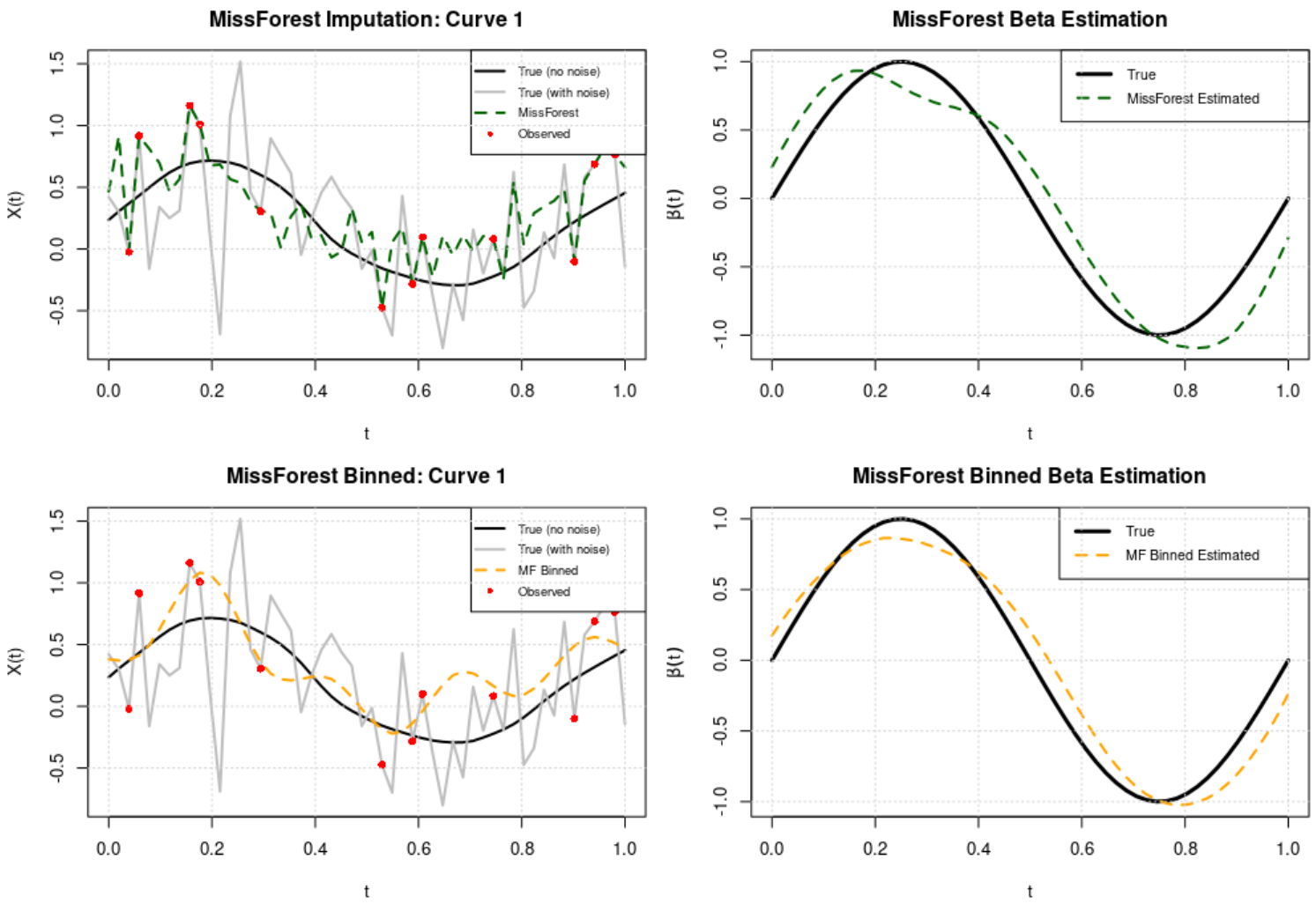

Figure: Results for MF and MF_B

Application

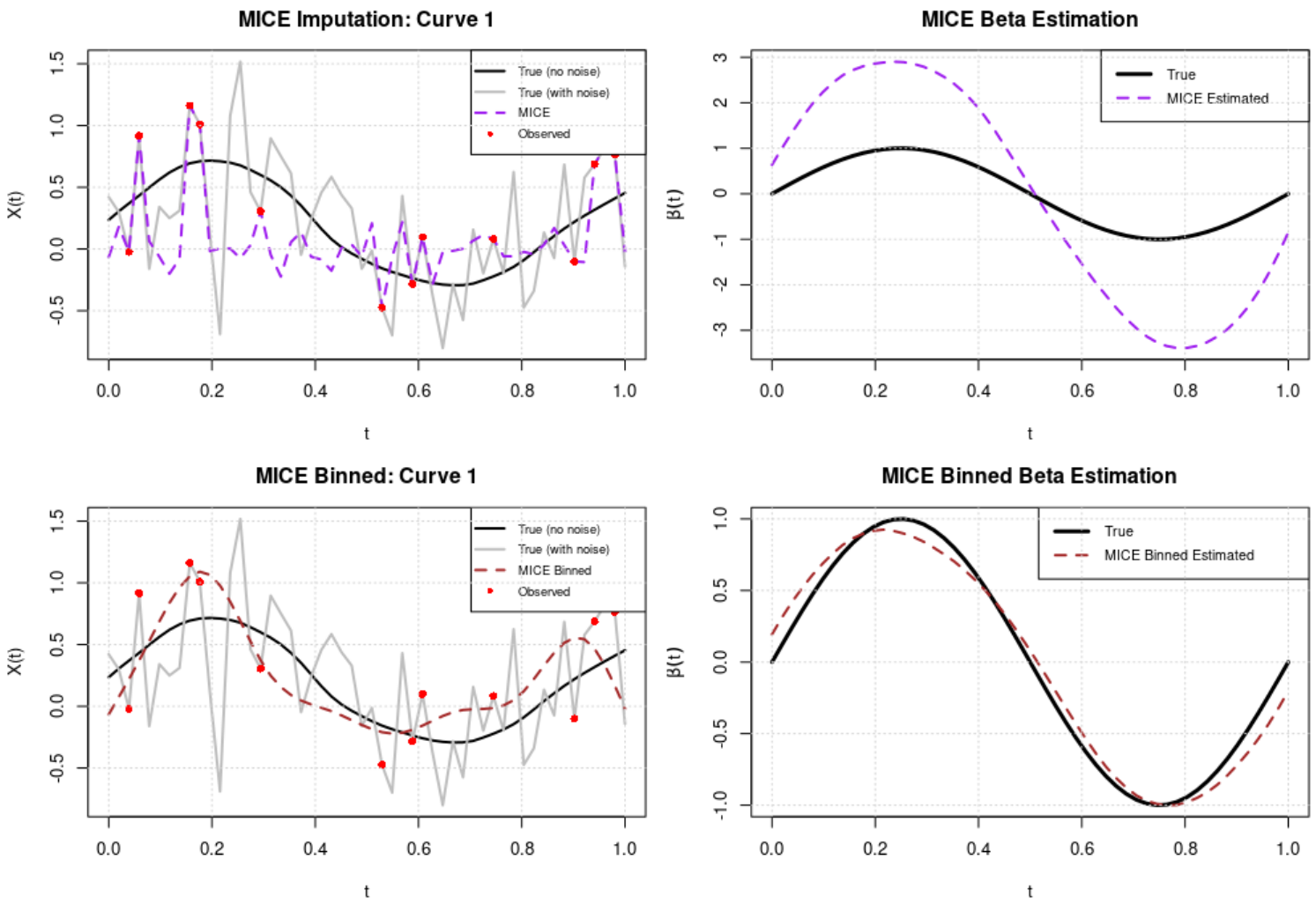

Figure: Results for MICE and MICE_B

Discussion

Some limitations of the paper:

- Imputation and analysis are performed in two separate steps, after which the imputed curves are treated as **fully observed**.
- Although **multiple imputation** is motivated as a way to reflect **uncertainty**, this uncertainty is not explored in the subsequent analysis.
- It is **unclear** whether the response variable Y is included during imputation in **MF** and **MICE**, even after consulting the original sources.

Discussion

Details that limit reproducibility:

- Several important implementation choices are **not described** in sufficient detail.
- Simulation settings, the scalar-on-function regression procedure, and hyperparameter choices are **missing** or only briefly mentioned.
- For the regression step, methods from Kokoszka and Reimherr, 2017 are **mentioned**, but it is **not stated** which specific method is actually used.

Conclusion

- **Tree-based methods** (MF, LLF) are flexible and handle complex dependencies, but may produce rough curves.
- **Binning** is a key enhancement: it stabilizes imputation and restores smooth functional structure.
- However, once binning is applied, MF_B performs similarly to the newly proposed MLLF_B in many settings.
- Exploring different **binning strategies** can perhaps improve performance even more.

Sources

- Kokoszka, P., & Reimherr, M. (2017). *Introduction to functional data analysis*. Chapman; Hall/CRC.
- Rao, A. R., & Reimherr, M. (2020).Modern multiple imputation with functional data. *Stat*, *10*(1), e331. https://doi.org/https://doi.org/10.1002/sta4.331

Thank you for your attention!

Are there any questions?